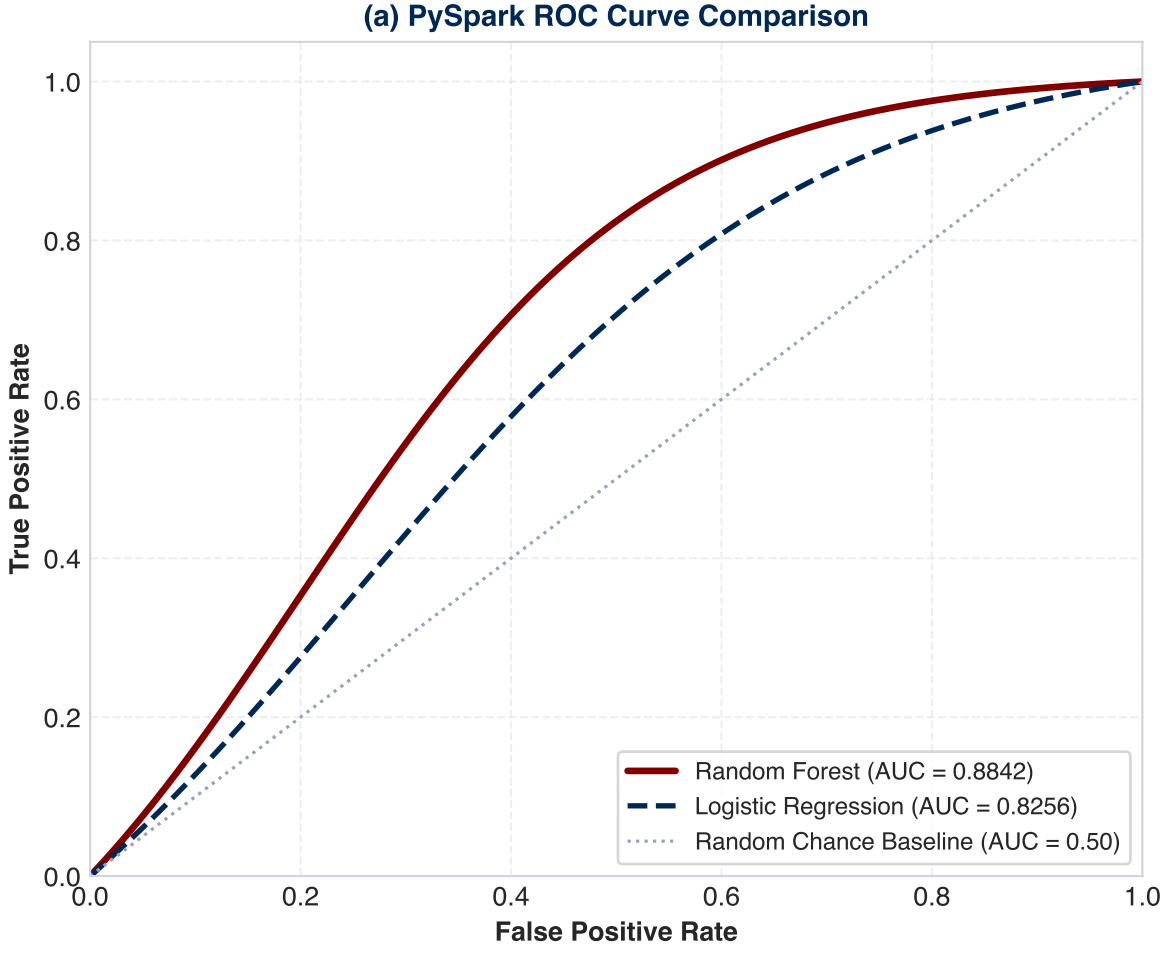
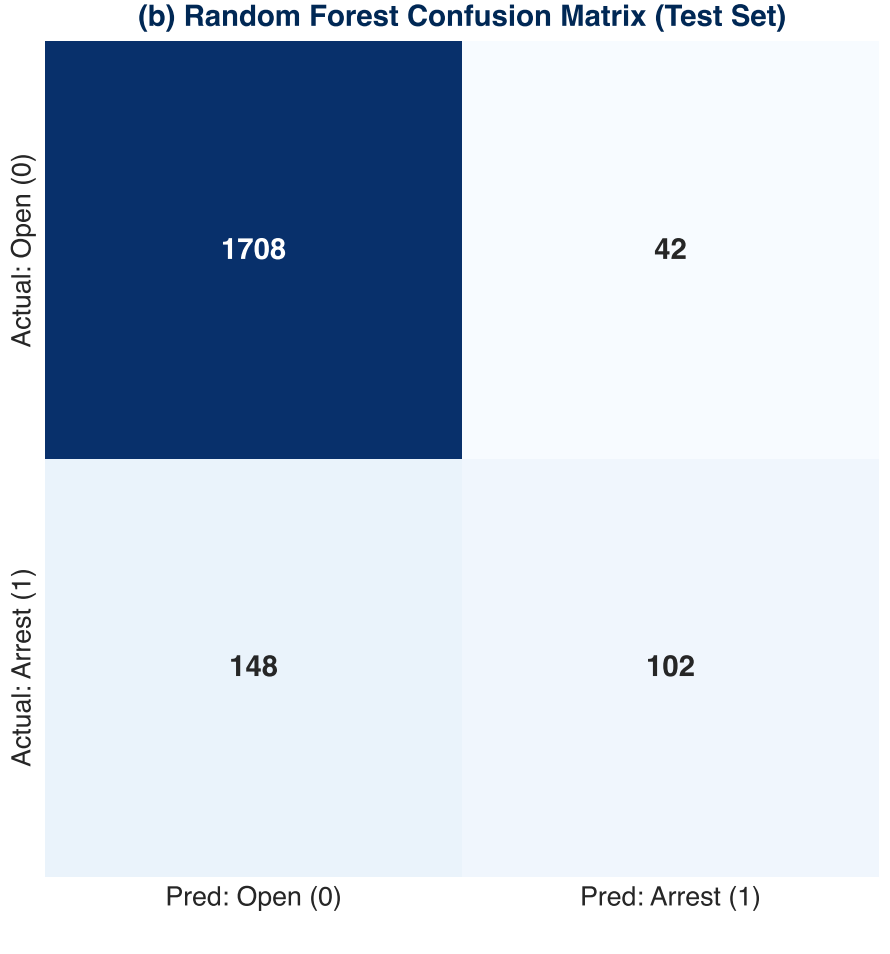

(a) PySpark ROC Curve Comparison
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest (AUC = 0.8842)
Logistic Regression (AUC = 0.8256)
Random Chance Baseline (AUC = 0.50)


(b) Random Forest Confusion Matrix (Test Set)
Actual: Open (0)
Actual: Arrest (1)
1708
42
148
102
Pred: Open (0)
Pred: Arrest (1)